

2(a)(i)	metaphase ;	1
2(a)(ii)	4 chromosomes within dotted line of both newly forming cells and each chromosome consisting of one chromatid ; ignore shape / shading labels not required centromeres do not need to be shown two obvious pairs (of chromosomes) shown in each of the two re-forming nuclei ; e.g. two dark and two light two longer and two shorter	2
2(b)	<i>any one from:</i> prevent the loss of, genes / genetic information ; A prevent loss of coding DNA AVP ; e.g. prevents the ends of two chromosome from fusing / joining	1
2(c)	<i>any three from:</i> 1 stem cells (grown in the lab) can undergo mitosis, ; A divide 2 stem cells are, undifferentiated / unspecialised ; 3 (when back in the person being treated) stem cells can, differentiate / specialise, into (all) <u>blood cell</u> types ; A are multipotent 4 to, re-populate / replace / bring back to normal levels, the person's own <u>blood cell</u> supply ; A tissue repair	3